

Far-Left Stability of a Chemotaxis Front

A mathematical proof of the sharp threshold $\chi = 4$, with the necessary basin condition

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Abstract

For the normalized one-dimensional chemotaxis–logistic equation

$$u_t = u_{zz} + (c - \chi v_z)u_z + u(1 - \chi v + (\chi - 1)u), \quad -v_{zz} + v = u,$$

we identify the exact stability threshold of the plateau $(u, v) = (1, 1)$. The linear dispersion relation is strictly stable exactly for $0 \leq \chi < 4$. A nonlinear logarithmic entropy has the same sharp coercive coefficient $1 - \chi^2/16$. A slowly varying integrable weight turns this identity into a whole-line Liouville theorem, and a translation-compactness argument gives uniform far-left convergence for orbits that remain in a persistent positive far-left band.

The band hypothesis is necessary for a broad class of front-like solutions. A Fisher–KPP traveling front observed from a faster frame remains positive at the left end of every fixed time slice, yet converges to zero at each fixed point of that frame. Thus slice-by-slice positivity cannot replace a time-uniform basin or phase condition.

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1 Equation and far-left convergence

In the normalized case $m = \alpha = \gamma = 1$, write the equation in a frame moving at speed c :

$$u_t = u_{zz} + cu_z - \chi(uv_z)_z + u(1 - u), \quad (1)$$

$$-v_{zz} + v = u. \quad (2)$$

Putting $r = v - 1$, equation (2) becomes

$$-r_{zz} + r = u - 1.$$

Since $v_{zz} = v - u$, expanding the chemotactic divergence in (1) gives the equivalent equation

$$u_t = u_{zz} + (c - \chi v_z)u_z + u(1 - \chi v + (\chi - 1)u).$$

A traveling front has the end states

$$(u, v) = (1, 1) \quad \text{as } z \rightarrow -\infty, \quad (u, v) = (0, 0) \quad \text{as } z \rightarrow +\infty.$$

The desired far-left conclusion for a time-dependent orbit is

$$\forall \varepsilon > 0 \exists R, T \forall t \geq T \forall z \leq -R, \quad |u(t, z) - 1| < \varepsilon. \quad (3)$$

2 The exact linear threshold

Set $u = 1 + p$, $v = 1 + r$. The linearized equations are

$$p_t = p_{zz} + cp_z + (\chi - 1)p - \chi r, \quad -r_{zz} + r = p.$$

For $p = e^{\lambda t + i k z}$ and $s = k^2$, the elliptic equation gives $\hat{r} = \hat{p}/(1 + s)$, hence

$$\lambda(s) = -s + i c k + (\chi - 1) - \frac{\chi}{1 + s}. \quad (4)$$

The real growth rate is

$$d_\chi(s) = \operatorname{Re} \lambda(s) = -1 - s + \frac{\chi s}{1 + s}. \quad (5)$$

Theorem 1 (sharp dispersion threshold). *For $0 \leq \chi < 4$, $d_\chi(s) < 0$ for every $s \geq 0$. At $\chi = 4$, the modes $k = \pm 1$ are neutral. For $\chi > 4$, a nonempty band of Fourier modes grows exponentially.*

Proof. If $0 \leq \chi \leq 1$, then

$$d'_\chi(s) = -1 + \frac{\chi}{(1 + s)^2} \leq 0,$$

so the maximum is $d_\chi(0) = -1$. If $\chi > 1$, the unique critical point is $s_* = \sqrt{\chi} - 1$, and $d''_\chi(s) < 0$, so it is the global maximum. Substitution gives

$$d_\chi(s_*) = \chi - 2\sqrt{\chi}.$$

This is negative, zero, or positive according as $\chi < 4$, $\chi = 4$, or $\chi > 4$. At equality $s_* = 1$, hence $k = \pm 1$. $\square$

More generally, a reaction exponent $\alpha > 0$ and coupling $\chi\gamma \geq 0$ give

$$d_{\alpha, \chi\gamma}(s) = -\alpha - s + \frac{\chi\gamma s}{1 + s}.$$

When $\chi\gamma > 1$, its maximum is $(\sqrt{\chi\gamma} - 1)^2 - \alpha$. The exact stability condition is

$$\chi\gamma < (1 + \sqrt{\alpha})^2.$$

3 The sharp nonlinear entropy

For $u > 0$, define

$$h(u) = u - 1 - \log u, \quad h'(u) = 1 - \frac{1}{u},$$

and abbreviate

$$A = \frac{u_z}{u}, \quad W = u - 1.$$

Lemma 1 (unweighted entropy identity). *On a periodic interval, or on the whole line when all boundary terms vanish,*

$$\frac{d}{dt} \int h(u) = - \int A^2 + \chi \int Ar_z - \int W^2. \quad (6)$$

Proof. Multiplying (1) by $h'(u)$ gives

$$\int h'(u) u_{zz} = - \int h''(u) u_z^2 = - \int A^2.$$

The constant drift is the integral of a derivative. For the chemotactic term,

$$\begin{aligned} -\chi \int h'(u) (uv_z)_z &= -\chi \int \left(1 - \frac{1}{u}\right) (u_z r_z + u r_{zz}) \\ &= -\chi \int (W_z r_z + W r_{zz} - Ar_z) = \chi \int Ar_z. \end{aligned}$$

The first two terms cancel after integration by parts. Finally,

$$\int h'(u) u(1 - u) = \int (u - 1)(1 - u) = - \int W^2.$$

□

Lemma 2 (sharp elliptic resolver estimate). *If $-r_{zz} + r = W$, then*

$$\int r_z^2 \leq \frac{1}{4} \int W^2. \quad (7)$$

The constant 1/4 is optimal.

Proof. Taking Fourier transforms gives

$$\widehat{r_z}(k) = \frac{ik}{1 + k^2} \widehat{W}(k).$$

By Plancherel,

$$\int r_z^2 = \int \frac{k^2}{(1 + k^2)^2} \left| \widehat{W}(k) \right|^2 dk.$$

The multiplier is at most 1/4, with equality when $k^2 = 1$. □

Young's inequality gives

$$\chi Ar_z \leq A^2 + \frac{\chi^2}{4} r_z^2.$$

Combining this with (6) and (7) yields

$$\frac{d}{dt} \int h(u) \leq - \left(1 - \frac{\chi^2}{16}\right) \int (u - 1)^2. \quad (8)$$

No smallness assumption on $u - 1$ was used. The nonlinear entropy margin is positive exactly for $|\chi| < 4$, matching the linear threshold.

4 Slow localization on the whole line

The unweighted entropy need not be integrable for a bounded entire solution. For $0 < k < 1$ and a center x_0 , introduce

$$\phi_{k,x_0}(x) = \exp\left[-k\sqrt{1 + (x - x_0)^2}\right]. \quad (9)$$

Then $\phi > 0$, $\phi \in L^1(\mathbb{R})$, and

$$|\phi_x| \leq k\phi.$$

4.1 A near-sharp weighted resolver bound

Let

$$\begin{aligned} S &= \int \phi W^2, & A_0 &= \int \phi r^2, & I &= \int \phi r_z^2, \\ C_0 &= \int \phi r_{zz}^2, & J &= \int \phi_x r r_z, & P &= \int \phi W r. \end{aligned}$$

Testing $-r_{zz} + r = W$ by ϕr and expanding the square of the equation give

$$A_0 + I + J = P, \quad (10)$$

$$2P \leq S + A_0, \quad (11)$$

$$S = A_0 + 2I + 2J + C_0. \quad (12)$$

Moreover,

$$|J| \leq \frac{k}{2}(A_0 + I),$$

and the nonnegativity of $\int \phi(r + r_{zz})^2$ gives

$$2I + 2J \leq A_0 + C_0.$$

Lemma 3 (weighted resolver). *For $0 \leq k < 1$,*

$$I \leq G(k)S, \quad G(k) = \frac{1+k}{4(1-k)}. \quad (13)$$

Proof. Equations (10)–(11) and the lower bound on J give

$$(1-k)(A_0 + I) \leq S.$$

Equation (12) and $2I + 2J \leq A_0 + C_0$ give

$$4I + 4J \leq S.$$

Hence

$$4I \leq S + 4|J| \leq S + 2k(A_0 + I).$$

Multiplying by $1-k$ and using $(1-k)(A_0 + I) \leq S$ yields

$$4(1-k)I \leq (1+k)S,$$

which is (13). □

The gain $G(k)$ tends to the sharp value $1/4$ as $k \downarrow 0$.

4.2 The explicit localization loss

Assume that $u \geq \ell > 0$. The elementary inequality

$$h(u) \leq \frac{(u-1)^2}{u} \leq \frac{(u-1)^2}{\ell} \quad (14)$$

follows from $1 - u^{-1} \leq \log u$. Weighted integration by parts, reserving a fraction k of the diffusion to control the weight flux, gives

$$\frac{d}{dt} \int_{\mathbb{R}} \phi h(u) \leq -(1 - \mathcal{L}(\chi, c, \ell, k)) \int_{\mathbb{R}} \phi (u-1)^2, \quad (15)$$

where

$$\mathcal{L}(\chi, c, \ell, k) = \frac{\chi^2 G(k)}{4(1-k)} + \frac{k}{4} + \frac{\chi k}{2}(1 + G(k)) + \frac{|c| k}{\ell}. \quad (16)$$

For completeness, the two estimates containing all weight losses are

$$\begin{aligned} \chi X &\leq (1-k)D + \frac{\chi^2}{4(1-k)}I, \\ F &\leq kD + \left(\frac{k}{4} + \frac{\chi k}{2} + \frac{|c| k}{\ell} \right) S + \frac{\chi k}{2}I. \end{aligned}$$

Here

$$D = \int \phi \left(\frac{u_z}{u} \right)^2, \quad X = \int \phi \frac{u_z}{u} r_z, \quad S = \int \phi (u-1)^2,$$

and F is the sum of terms in which a derivative falls on ϕ . The first inequality is Young's inequality. The second follows from $|\phi_x| \leq k\phi$, (14), and Young's inequality on each diffusion and chemotactic flux. Inserting $I \leq G(k)S$ gives (15)–(16).

At $k = 0$,

$$\mathcal{L}(\chi, c, \ell, 0) = \frac{\chi^2}{16}.$$

The loss is continuous at $k = 0$. Therefore, for every $0 \leq \chi < 4$, every $c \in \mathbb{R}$, and every $\ell > 0$, one can choose

$$0 < k < 1, \quad \mathcal{L}(\chi, c, \ell, k) < 1. \quad (17)$$

5 A whole-line Liouville theorem

Theorem 2 (entropy rigidity). *Let (u, r) be a bounded classical solution on all $(t, x) \in \mathbb{R}^2$ of*

$$u_t = u_{xx} + cu_x - \chi(ur_x)_x + u(1-u), \quad -r_{xx} + r = u-1.$$

Assume $0 \leq \chi < 4$ and

$$0 < \ell \leq u(t, x) \leq M \quad \text{for all } (t, x) \in \mathbb{R}^2,$$

together with the regularity and boundedness needed for the weighted integrations above. Then $u \equiv 1$ and $r \equiv 0$.

Proof. Choose k as in (17), fix x_0 , and define

$$E(t) = \int_{\mathbb{R}} \phi_{k,x_0}(x) h(u(t,x)) dx.$$

The bounded range and $\phi \in L^1$ give a uniform bound

$$0 \leq E(t) \leq K \quad (t \in \mathbb{R}).$$

By (14) and (15),

$$E'(t) \leq -\rho E(t), \quad \rho = (1 - \mathcal{L}(\chi, c, \ell, k))\ell > 0.$$

Starting at $t - T$ and applying Gronwall's inequality gives

$$0 \leq E(t) \leq E(t - T)e^{-\rho T} \leq Ke^{-\rho T}.$$

Letting $T \rightarrow \infty$ yields $E(t) = 0$. Since $\phi > 0$, $h(u) \geq 0$, and $h(u) = 0$ only at $u = 1$, continuity implies $u(t, x) = 1$ for every t, x . The elliptic equation then gives the bounded solution $r \equiv 0$. $\square$

The use of negative time is legitimate because the theorem concerns an entire translated limit. This is the decisive advantage of combining entropy with late-time translation compactness.

6 Far-left convergence by compactness

We state the remaining analytic input without any hidden notation. Assume:

(H1) there are T, Z, ℓ, M , with $\ell > 0$, such that

$$\ell \leq u(t, z) \leq M \quad (t \geq T, z \leq Z); \quad (18)$$

(H2) for every pair of sequences $t_n \rightarrow +\infty$ and $z_n \rightarrow -\infty$, the translates

$$u_n(t, x) = u(t + t_n, x + z_n), \quad r_n(t, x) = r(t + t_n, x + z_n)$$

and the fields

$$(u_n)_x, (u_n)_{xx}, (u_n)_t, r_n, (r_n)_x, (r_n)_{xx}$$

are locally equicontinuous. Moreover, there is a constant C , independent of the chosen sequences and of the compact box, such that all seven fields have absolute value at most C on each fixed compact box for all sufficiently large n . Consequently a common subsequence converges locally uniformly to globally bounded fields, with the differential identities passing to the limit.

Theorem 3 (basin-conditional far-left convergence). *If $0 \leq \chi < 4$ and (H1)–(H2) hold, then (3) holds.*

Proof. Suppose (3) fails. Then there are $\varepsilon_0 > 0$, $t_n \rightarrow \infty$, and $z_n \rightarrow -\infty$ such that

$$|u(t_n, z_n) - 1| \geq \varepsilon_0.$$

By (H2), a subsequence of the translates converges locally uniformly to a bounded entire classical solution (U, R) . Hypothesis (18) passes to the limit and gives $\ell \leq U \leq M$ on all of $\mathbb{R}^2$. The differential equations also pass to the limit. The Liouville theorem therefore gives $U \equiv 1$. On the other hand, local convergence at the origin gives

$$|U(0, 0) - 1| = \lim_n |u(t_n, z_n) - 1| \geq \varepsilon_0,$$

a contradiction. $\square$

7 Why the basin condition is necessary

Slice-by-slice left positivity has the quantifier pattern

$$\forall t \exists \ell_t > 0 \exists Z_t \forall z \leq Z_t, \quad u(t, z) \geq \ell_t.$$

The persistent band (18) instead requires

$$\exists \ell > 0 \exists T, Z \forall t \geq T \forall z \leq Z, \quad u(t, z) \geq \ell.$$

The order of the quantifiers cannot be interchanged.

Proposition 1 (convective escape). *At $\chi = 0$, there is a genuine global classical front which is positive at the left end of every fixed time slice but has no persistent positive floor when observed from a sufficiently fast frame.*

Proof. For every $s \geq 2$, the Fisher–KPP equation has a monotone traveling front U_s satisfying

$$U_s'' + sU_s' + U_s(1 - U_s) = 0, \quad U_s(-\infty) = 1, \quad U_s(+\infty) = 0.$$

Take $s = 3$, and let V_s be the bounded solution of $-V_s'' + V_s = U_s$. In a frame of speed $c = 4$, set

$$q(t, z) = U_s(z + (c - s)t), \quad w(t, z) = V_s(z + (c - s)t).$$

The chain rule and the wave equation give

$$q_t = q_{zz} + cq_z + q(1 - q), \quad -w_{zz} + w = q,$$

which is precisely the $\chi = 0$ moving-frame system. For each fixed t , $q(t, z) \rightarrow 1$ as $z \rightarrow -\infty$. But for each fixed z ,

$$z + (c - s)t \rightarrow +\infty, \quad q(t, z) \rightarrow 0.$$

Given any proposed $\ell > 0, T, Z$, choose $t \geq T$ so large that $q(t, Z) < \ell$. Thus no persistent band exists. $\square$

This counterexample does not contradict stability relative to a fixed wave under a weighted closeness or phase condition: such a condition pins the solution to the same wave speed. It rules out only an unconditional far-left theorem over arbitrarily phased front-like orbits.

8 Conclusion

The number 4 is sharp in three compatible senses:

1. below 4, every Fourier mode decays; at 4, $k^2 = 1$ is neutral; above 4, a band of modes grows;
2. the nonlinear entropy has the exact coefficient $1 - \chi^2/16$;
3. throughout $0 \leq \chi < 4$, slow localization, the entire-orbit Liouville theorem, and translation compactness yield uniform far-left convergence once a persistent positive band is supplied.

The band or an equivalent wave-phase condition is structural, not technical: without it, a genuine front can leave a faster observation frame by convective escape.